Task 6 — SQL Script

```
-- Task 6: Sales Trend Analysis Using Aggregations
-- Compatible with SQLite.
-- Dataset/table: online_sales
-- Expected columns: order_date, amount, product_id, order_id

-- 1. Monthly revenue and order volume
SELECT
CAST(strftime('%Y', order_date) AS INTEGER) AS year,
CAST(strftime('%m', order_date) AS INTEGER) AS month,
ROUND(SUM(amount), 2) AS monthly_revenue,
COUNT(DISTINCT order_id) AS order_volume
FROM online_sales
WHERE order_date IS NOT NULL
GROUP BY year, month
ORDER BY year, month;

-- 2. Top 3 months by sales/revenue
SELECT
CAST(strftime('%Y', order_date) AS INTEGER) AS year,
CAST(strftime('%m', order_date) AS INTEGER) AS month,
ROUND(SUM(amount), 2) AS monthly_revenue
FROM online_sales
WHERE order_date IS NOT NULL
GROUP BY year, month
ORDER BY monthly_revenue DESC
LIMIT 3;

-- 3. Example: results for a specific time period
-- Change the dates as required.
SELECT
CAST(strftime('%Y', order_date) AS INTEGER) AS year,
CAST(strftime('%m', order_date) AS INTEGER) AS month,
ROUND(SUM(amount), 2) AS monthly_revenue,
COUNT(DISTINCT order_id) AS order_volume
FROM online_sales
WHERE order_date >= '2024-01-01'
AND order_date < '2025-01-01'
GROUP BY year, month
ORDER BY year, month;
```